

Generating and Comparing Two Patterns

Generating terms, relating the rules, comparing terms, and turning patterns into ordered pairs

Directions

- Problems 1 to 5 and problem 10 use the two rules in the table below. Both patterns start at 0 and neither ever stops.
- State the terms you generated before answering. Include the unit whenever the problem gives one.

Term number	1	2	3	4	5	6
Pattern A — start at 0, add 4	0	4	8			
Pattern B — start at 0, add 8	0	8	16			

Two Rules Table — fill in the blanks as you work; problems 1 to 5 and 10 use it.

1. Write the 6th term of pattern A.

Answer: _____

2. Write the 6th term of pattern B.

Answer: _____

3. Compare the two rules. Each time pattern A adds its step, pattern B adds a step how many times as large?

Answer: _____

4. How much greater is the 10th term of pattern B than the 10th term of pattern A?

Answer: _____

5. Write the ordered pair (pattern A's 4th term, pattern B's 4th term).

(_____ , _____)

6. Maya has 12 dollars saved and adds 3 dollars every week. Liam has 4 dollars saved and adds 5 dollars every week. Neither of them spends anything. After how many weeks will they have the same amount?

Answer: _____ weeks

7. How many dollars will Maya and Liam each have at the moment their savings are equal?

Answer: _____

8. The table below gives five terms of pattern P and five terms of pattern Q.

Term number	1	2	3	4	5
Pattern P	7	10	13	16	19
Pattern Q	2	6	10	14	18

Write the ordered pair (pattern Q's 5th term, pattern P's 5th term).
(_____ , _____)

9. Use the table of patterns P and Q from problem 8.

(a) Write the mean of pattern P's five terms.

Answer: _____

(b) Write the mean of pattern Q's five terms.

Answer: _____

10. Return to patterns A and B from the Two Rules Table.

(a) Write the 12th term of pattern A.

Answer: _____

(b) Write the 12th term of pattern B.

Answer: _____

(c) Describe the relationship between matching terms of the two patterns, and explain why that relationship must hold for every term, not just the ones you checked.
